

Component 3: Monthly Temperature Request

xPostForecast · Milestone 4: Component-Level Design · Pseudocode

Where it runs	Backend: GET /temperature/:date (backend/middleware/authMiddleware.js, backend/routes/stac.js)
User stories	US-3, US-6, US-8
Flowchart	Component3_MonthlyTemperatureRequest_Flowchart.pdf

Description

Answers a logged-in user's request for one month of West Virginia temperatures: checks the session, validates the date, and returns the month's points from the cache or from Component 4.

Design principles

Separation of concerns. Checking the session is a cross-cutting concern, so it lives in its own module, AuthGuard, which runs before the request handler. The temperature logic never deals with cookies or tokens, and the same guard can protect any future route.

Modularity. Fetching data from the outside service is a separate module, LoadMonth (Component 4), shown in the flowchart as a predefined process. This component only decides what to do with LoadMonth's three possible outcomes (points, no data, or failure), which keeps each module small enough to understand and test on its own.

Pseudocode

```
BEGIN GetMonthlyTemperatures
  READ token FROM the "token" cookie

  IF token is missing THEN
    RESPOND 401 "Not logged in"
    STOP
  END IF

  IF the token's signature is invalid OR the token has expired THEN
    RESPOND 401 "Invalid or expired session"
    STOP
  END IF

  READ date FROM the request path

  IF date is not in the form YYYY-MM-DD with a month from 01 to 12 THEN
    RESPOND 400 "Date must be YYYY-MM-DD"
    STOP
  END IF

  SET month TO the year and month of date (YYYY-MM)

  IF the cache contains month THEN
    RESPOND 200 WITH the cached points
    STOP
  END IF
```

```
CALL LoadMonth(month), GIVING points          (Component 4)

IF LoadMonth failed THEN
    LOG the error
    RESPOND 502 "The temperature data service is unavailable. Try again later."
    STOP
END IF

IF points is NONE THEN
    RESPOND 404 "No temperature data is available for <month>"
    STOP
END IF

STORE points IN the cache under month

IF the cache holds more than 24 months THEN
    REMOVE the oldest month from the cache
END IF

RESPOND 200 WITH points
END GetMonthlyTemperatures
```